INCH-POUND

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SUPERSEDING
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3 November 1986

MILITARY SPECIFICATION

DRILLING MACHINE, UPRIGHT, BOX COLUMN, FLOOR TYPE PLAIN TABLE, HAND FEED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers electric motor driven, floor mounted, box column, sliding head, vertical, manual feed, single spindle, and gang drilling machines, with accessories as described herein.

1.2 Classification. The drilling machines shall be of the following styles and sizes. The style and size of the machine to be furnished shall be as specified (see 6.2).

Style A - One spindle
Style B - Two spindle
Style C - Three spindle
Style D - Four spindle
Style E - Five spindle
Style F - Six spindle

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, 2163 Airways Boulevard, Memphis, Tennessee 38114-5051, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3413

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Size	Drill capacity (inches)	Style A Swing (inches)	Styles B through F overhang (inches)
12	5/16	12	6
14	3/8	14	7
16	1/2	16	8
18	3/4	18	9
20	1	20	10
24	1-1/4	24	12

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interferences.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

(Unless otherwise indicated, copies of Federal and military specifications, standards, and handbooks are available from the Standardization Document Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19120-5099.)

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2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

U. S. DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

29 CFR 1910 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402-0001.)

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B5.1 - T-Slots, Their Bolts, Nuts, and Tongues. (DOD adopted)

ANSI B5.10 - Machine Tapers. (DOD adopted)

ANSI B11.8 - Drilling, Milling, and Boring Machines, Safety Requirements for the Construction, Care, and Use. (DOD adopted)

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Department, 1430 Broadway, New York, NY 10018-3363.)

NATIONAL ELECTRICAL MANUFACTURERS' ASSOCIATION (NEMA)

NEMA MG 1 - Motors and Generators. (DOD adopted)

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(Application for copies should be addressed to the National Electrical Manufacturers' Association, 2101 L Street, N.W., Washington, DC 20037-1580.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Industrial Machinery. (DOD adopted)

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), one complete drilling machine shall be subjected to first article inspection (see 6.4) in accordance with 4.4.

3.2 Design. The drilling machine shall be new (not a prototype) and one of the manufacturer's current production models capable of operations in accordance with the requirements herein. The drilling machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be readily accessible for adjustment, replacement, or repair.

3.2.1 Measuring and indicating device calibrations. Measuring and indicating devices shall be graduated in inch-pound (see 6.5.1), metric (see 6.5.2), or dual (inch-pound and metric) units as specified by the procuring activity (see 6.2). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.1.1 Dials. All dials used to indicate machine axis position or tool movement shall be graduated in increments not larger than 0.001 inch. Dial and handwheel circumferences shall be permanently and legibly engraved or etched on a nonglare background with graduations that can be read from the operator's normal position. All feed dials shall have

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independent zero adjustments and shall be calibrated in such a manner that the last dial graduation progresses into and is continuous with the first dial graduation as the dial is rotated through the zero position.

3.2.2 Reclaimed materials. The machine may contain reclaimed materials to the maximum extent possible provided such materials will not jeopardize the intended use, performance, or design life of the machine. Reclaimed materials shall have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used, rebuilt, or remanufactured end products are allowed under this specification.

3.2.3 Energy efficiency. The drilling machine and its components that directly consume energy in normal operation shall be designed and constructed for energy efficiency in accordance with the developments available within the industry.

3.2.4 Controls. All mechanical, hydraulic, and pneumatic operating controls shall be located convenient to the operator's work station(s).

3.2.5 Safety and health requirements. All parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall conform to all requirements of OSHA 29 CFR 1910 and ANSI B11.8. If a conflict arises between 29 CFR 1910 and ANSI B11.8, 29 CFR 1910 shall apply. Additional safety and health requirements shall be as specified (see 6.2). Covers, guards, or other safety devices normally furnished as standard on the manufacturer's commercial drilling machine supplied to the commercial market shall be provided for the point of operation and all other parts of the drilling machine that present safety hazards. If point-of-operation guarding is influenced by workpiece size and configuration, the procuring activity shall provide part design data to the supplier of the machine (see 6.2). The supplier of the machine shall provide the special guarding which shall meet OSHA 29 CFR 1910.

3.2.6 Mercury restriction. The drilling machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the drilling machine.

3.2.8 Environmental protection. During the manufacture, operation, service, transportation, or storage of the machine and its components, materials hazardous to the ecological system as prohibited by applicable Federal, state, or local statutes in effect on the date of the contract shall not be used or emitted.

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3.2.9 Lubrication. Means shall be provided to ensure sufficient lubrication for all moving parts. All oil holes, grease fittings, and filler caps shall be readily accessible.

3.2.10 Interchangeability. To provide for replacement of worn parts, all parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or machine. Replacement parts shall be domestically available.

3.3 Construction. The drilling machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the drilling machine to its maximum rating and capacity without permanent distortion or failure.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Fastening devices. All fasteners shall be installed to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be permanently installed.

3.3.3 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.4 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. Any material used for such operations shall be thoroughly removed from the part(s) upon completion of the operations.

3.3.5 Painting. Unless otherwise specified (see 6.2), the paint color and application shall be in accordance with the manufacturer's commercial practice.

3.3.6 Threads. All threaded parts used on the drilling machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Gears. All gears shall be constructed from a material suitable for the intended purpose. All metal gears shall be machined and heat

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treated by a process that will impart the hardness, surface finish, and toughness that will enable the gear train to transmit full-rated torque to the drive motor without gear damage, failure or premature wear.

3.3.8 Electromagnetic interference control. When specified (see 6.2), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystem(s) class and the emission and susceptibility requirements shall be as specified.

3.4 Components.

3.4.1 Base. The base shall be a rib-braced casting of iron or iron alloy, or a steel weldment, and shall provide a foundation for the entire drilling machine. Means shall be provided for fastening the machine to the floor. When a coolant system is required under 3.4.10, the sump or reservoir may be cast as an integral part of the base or furnished as a separate external unit. One or more screw jacks for table support and height adjustment shall be mounted on the base. Gang drills shall have a common base arranged to accept a common table and shall have the number of spindles and all of their related components for the style machine required.

3.4.2 Column. The column shall be a casting of iron or iron alloy, or a steel weldment, reinforced to provide support and minimize deflection. The column shall be of box construction and shall be arranged to receive and carry the drill head assembly. Guideways shall be provided for accepting the sliding head and worktable. Vertical adjustments of both the sliding head and worktable shall be provided with means for locking. The column support surfaces shall be machined to accurately fit those of the base. Columns on gang drills shall be duplicates of single-spindle machines. When specified (see 6.2), gang drills shall be furnished with an adjusting type column. The adjusting mechanism shall allow the operator to change the center-to-center distance of the spindles on the machine base. Means for clamping after adjustment shall be provided.

3.4.3 Worktable. The worktable shall be a casting of iron or iron alloy, reinforced for support and deflection. Table height adjustment on the column shall be by a screw jack arrangement, with means for clamping after positioning. Worktables shall be rectangular with trough, having provisions for draining. When specified (see 6.2), a replacement type-wear plate, accurately fitted to the worktable top shall be provided. Work surfaces, of either the table or wear plate, shall be machined to not rougher than 75 microinch and be flat within 0.002 inch. When specified (see 6.2), T-slots shall be provided for either the worktable or wear plate, as applicable. The number of T-slots and their configuration shall be as specified. Machined T-slots shall be in accordance with ANSI B5.1.

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3.4.4 Drillhead assembly. The drillhead assembly shall consist of the drillhead, quill and spindle assembly, power transmission and sliding head, and when required by design, the upper portion of the column. Means for vertical travel of both the spindle and the sliding head shall be incorporated in the assembly design. The assembly shall be operable as a unit and be interchangeable among other machines of the same size and model.

3.4.5 Drillhead. The drillhead shall be designed to house the power transmission. When the drillhead is not an integral part of the column, it shall be held in position by clamps or other positive means.

3.4.6 Quill and spindle assembly. The quill shall be machined of heat-treated alloy steel, or cast steel. The quill shall house the spindle and be fitted to the sliding head. Manual drill feeds shall be accomplished by means of a hand-feed lever through a rack and pinion. A spring assembly or counterweight apparatus shall retract the quill and spindle assembly when it is released. The spring assembly shall be adjustable for tension. Quill guides shall be provided to support and maintain quill alignment through the full travel of the quill. The spindle shall be machined from heat-treated alloy steel and shall be mounted in the quill on ball or roller bearings. Unless otherwise specified (see 6.2), the driven end of the spindle shall be multi-splined. Unless otherwise specified, the spindle shall have a tapered socket hole. Taper shall be manufacturers' standard but shall conform to ANSI B5.10. When specified (see 6.2), one of the following spindle nose configurations shall be provided.

a. Shank type for drill chuck.

b. Threaded end for drill chuck.

c. Solid-male tapered type.

3.4.7 Power transmission assembly. The drilling machine shall be powered by an electric motor. The drive shall be through a belt and step pulley arrangement or belt and gears or all gear drive or direct motor spindle drive. Means for adjusting belt tension shall be provided when a belt-type drive is used. When specified (see 6.2), power feed shall be provided. The number of feeds and their feed range shall be as specified.

3.4.8 Sliding head. The sliding head shall be mounted on the column, and shall serve as a support for the quill and spindle. The head shall be vertically adjustable. The slides of the head shall accurately fit the ways of the column and shall be provided with means for clamping after adjustment.

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3.4.9 Depth stop. Each spindle of each machine shall be provided with a depth stop equipped with a positive locking device. When specified (see 6.2), the depth stop shall be provided with an indicating scale marked in increments of not more than 1/16 of an inch.

3.4.10 Coolant system. When specified (see 6.2), a coolant system shall be provided. The coolant system shall include a sump or reservoir, a power driven pump, and all necessary piping. The sump or reservoir shall have sufficient capacity to permit full flow of coolant. The system shall have means for draining and cleaning and shall include a baffle or a strainer which is easily removable for cleaning. Means shall be provided to permit the operator to direct and control the amount of coolant over the entire work area.

3.4.11 Electrical requirements. The machine shall be wired to conform to NFPA 79. Unless otherwise specified (see 6.2), each machine shall be designed to operate on dual 115/230-volts, single phase 60-Hertz (Hz). Unless otherwise specified (see 6.2), the machine shall be initially wired for operation on 230 volts, or when specified (see 6.2), machines with motors over 1-hp, 230/460 volts alternating-current (AC), 3-phase,60-Hz.

3.4.11.1 Motors. Motors shall be rated for a 30-minute duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Unless otherwise specified (see 6.2), alternating-current motors shall be designed to operate on 60-Hz. The motor enclosure shall meet the requirements for a totally enclosed fan-cooled enclosure. All motors shall conform to the requirements of NEMA MG-1.

3.4.11.2 Worklight. When specified (see 6.2), a worklight shall be provided. The light shall have a protective shield, an on-off switch, and shall be adjustable to reflect maximum light on the work area. If a built-in lighting system is provided, the adjustable light described above shall not be required.

3.4.11.3 Hour meter. When specified (see 6.2), the drilling machine shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the main drive motor. The meter shall be of the nonresetting type and shall have a range of 0 to 99,999 hours in increments of not greater than 1 hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the drilling machine.

3.5 Size and capacity. Unless otherwise specified (see 6.2), the size and capacity of the drilling machine shall meet the requirements of table I for the machine specified. Machine capacity shall be not less than the stated requirements of the size ordered. When a range is given, the required performance shall be from the stated minimum or less to the stated maximum or greater.

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3.5.1 Physical size limitations. When specified (see 6.2), the physical size of the machine shall not exceed the height, width, and length restrictions specified by the procuring activity to ensure the machine will fit in its future operating location.

3.6 Alignment tolerances. Each drilling machine shall meet the alignment tolerances specified in table II.

3.7 Performance. The drilling machine shall be capable of drilling 15 holes of the maximum capacity drill size through 1-inch thick soft cast iron within 20 minutes.

3.8 Standard equipment. Unless otherwise specified (see 6.2), the following equipment shall be furnished with each drilling machine:

a. One set of hand tools normally furnished by the manufacturer.

b. One drill chuck for each spindle, 5/16-inch capacity, having arbor taper corresponding to spindle taper for the size machine specified.

3.9 Optional equipment. Optional equipment shall be furnished as specified and fully described (see 6.2). The optional equipment provided shall be functional without requiring modification of the equipment or machine.

3.10 Repair parts. Required repair parts shall be furnished as specified (see 6.2).

3.11 Marking on instruments, control panels, charts, and plates. All words on instruments, control panels, charts, and plates shall be in the English language. Characters shall be permanently marked in boldface on a contrasting background. All plates shall be corrosion resistant.

3.11.1 Lubrication plate. When specified (see 6.2), a lubrication plate shall be attached to each machine. Unless otherwise specified (see 6.2), the information provided on the plate shall be as listed below.

Points of lubricant application
Servicing interval
Type of lubricant
Federal or military specification for each lubricant (if applicable)

3.11.2 Nameplate. A nameplate shall be securely attached to each machine. Unless otherwise specified (see 6.2), the nameplate shall contain the information listed below.

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Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Amp rating of largest motor
Short circuit/over-current protection rating
Contract number or order number
National Stock Number or Plant Equipment Code
Date of manufacture

3.12 Workmanship. Workmanship of the drilling machine and its accessories shall meet all requirements specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submissions of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

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4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the test conditions specified in 4.7.1.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.6 and all tests in 4.7. The machine shall pass the examination and all tests to be accepted.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each machine prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of the examination in 4.6, the tests in 4.7, and the inspection in 4.8. The machine shall pass the examination, all tests, and the inspection to be accepted.

4.6 Examination. Each drilling machine shall be examined to determine compliance with all requirements of this specification.

4.7 Tests.

4.7.1 Test conditions. All tests shall be performed in an indoor facility with ambient conditions of 41° to 104° Fahrenheit and 20 to 95 percent relative humidity.

4.7.2 Operational test. Each drilling machine shall be operated at no load for not less than 30 minutes. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.7.3 Alignment tolerance test. Each machine shall be tested to determine compliance with the alignment tolerance requirements of 3.6. The alignment tolerance test shall be performed prior to and subsequent to the performance test(s) given in 4.7.4. Any machine adjustments required to comply with 3.6 during or after the performance test shall be cause for rejection.

4.7.4 Performance tests. The drilling machine shall be tested for conformance with 3.7. Any evidence of overloading, overheating, excessive vibration or deformation of the machine during or after drilling operations shall be cause for rejection. All performance tests shall be completed without making any adjustments to or performing any maintenance on the machine.

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4.7.5 Electromagnetic interference control test. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.8 using the procedures given in MIL-STD-462.

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-18058.

6. NOTES

6.1 Intended use. The drilling machines covered by this specification are intended for use in production shops where wide varieties of drilling operations are performed.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Style and size required (see 1.2).

c. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

d. First article, if required (see 3.1).

e. System of units for measuring and indicating devices (see 3.2.1).

f. Additional safety and health requirements, if required (see 3.2.5).

g. Specify configuration of additional point-of-operation guarding, if required (see 3.2.5).

h. Painting, if different (see 3.3.5).

i. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.8).

j. Adjusting type column, if required (see 3.4.2).

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k. Wear plate, if required (see 3.4.3).

l. Specify the number and configuration of T-slots, if required (see 3.4.3).

m. Specify when drive end of spindle is not required to be multi-spindle (see 3.4.6).

n. Specify type spindle nose configuration required for drill chuck, if different (see 3.4.6).

o. Power feed, if required, and feeds and range (see 3.4.7).

p. Depth stop marked indicator, if required (see 3.4.9).

q. Coolant system, if required (see 3.4.10).

r. Electrical requirements, if different (see 3.4.11).

s. Initial voltage, if different (see 3.4.11).

t. Motors, if different (see 3.4.11.1).

u. Worklight, if required (see 3.4.11.2).

v. Hour meter, if required (see 3.4.11.3).

w. Size and capacity, if different (see 3.5).

x. Specify physical size limitation, if required (see 3.5.1).

y. Standard equipment, if different (see 3.8).

z. Specify and fully describe optional equipment, if required (see 3.9).

aa. Specify applicable repair parts, if required (see 3.10).

bb. Lubrication plate, if required (see 3.11.1).

cc. Information to be provided on lubrication plate, if different (see 3.11.1).

dd. Nameplate, if different (see 3.11.2).

ee. Inspection conditions, if different (see 4.3).

ff. First article inspection, if different (see 4.4).

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gg. Quality conformance inspection, if different (see 4.5).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
3.13	DI-TMSS-80527	Commercial Off-The-Shelf (COTS) Manuals	Require all data be in the in the English Language
3.13	DI-TMSS-80528	Supplemental Data Commercial Off-The-Shelf (COTS) Manual	Require all be in the English Language

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.3.1 Technical data. It should be specified in the "Remarks" section of the DD Form 1423 that all technical data ordered shall be written in the English language.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

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6.5 Definitions.

6.5.1 Inch-pound units. Inch-pound units are a system of measures based on the yard and pound commonly used in the United States of America and defined by the National Institute of Standards and Technology. Inch-pound units having the same names in other countries may differ in magnitude.

6.5.2 Metric units. Metric units are a system of basic measures defined by the International System of Units based on "Le Systeme International D'Unites (SI)," of the International Bureau of Weights and Measures. These units are described in ASTM E 380 and IEEE 268 (MIL-STD-961).

6.6 Measurement system. In this specification, all measurements, dimensions, sizes, and capacities are given in inch-pound units. These measurements may be converted to metric units through the use of the conversion factors and methods specified in FED-STD-376.

6.7 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.8 Safety and health requirements. OSHA 29 CFR 1910 limits only the total hazard level (noise, radiation, electromagnetic emissions, noxious vapors, air contaminants, and heat) of the environment in which a machine will operate. It does not limit the hazard level of individual machines in an operating environment. The procuring activity should analyze the existing hazard level in the proposed operating environment and specify additional requirements necessary to integrate this new machine into its future environment.

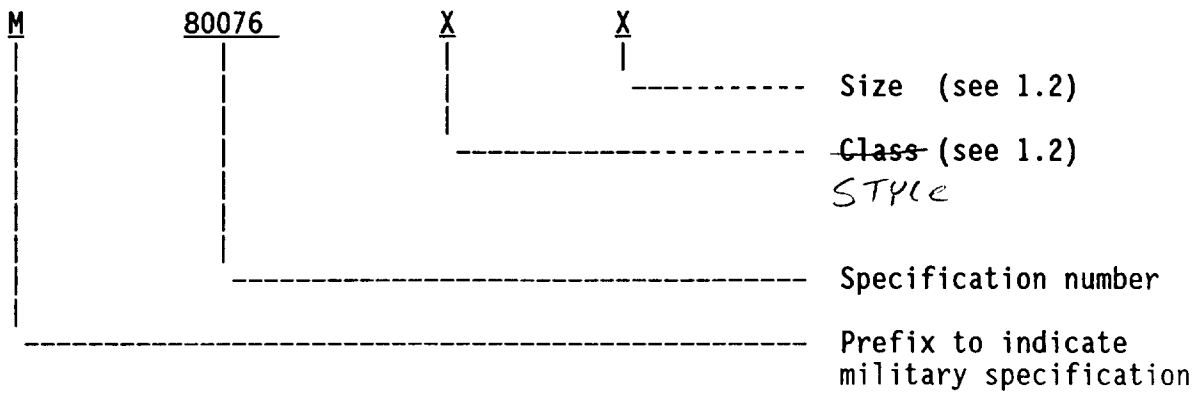
6.9 Training. Training required by the procuring activity should be provided as specified in the contract.

6.10 Warranty. Warranty requirements should be as specified by the procuring activity in the contract.

6.11 Inspection location. The contractor should identify in his response to the solicitation the location where inspection and tests are to be performed.

6.12 Part or Identifying Number (PIN). The PIN to be used for machines acquired to this specification are created as follows:

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6.13 Subject term (keyword) listing.

Floor type
Hand feed
Multiple spindle
Power feed
Single spindle
Vertical

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TABLE I - Size and capacities.

Characteristics	Size 12	Size 14	Size 16
Capacity (in mild steel), inches	5/16	3/8	1/2
Centerline of spindle to column face, inches	6	7	8
Swing, inches, Style A	12	14	16
Overhang, inches, Style B through F	6	7	8
Sliding head adjustment, inches	3-1/2	3-1/2	4
Spindle traverse, inches	3	3	4
Spindle nose to table maximum inches	16	20	21
Spindle speeds, number	Infinite	Infinite	Infinite
Spindle speed range, revolution per minute	625 to 5000	625 to 5000	500 to 4000
Spindle diameter, (at drive end), inches	1/2	19/32	3/4
Quill diameter, inches	1-3/4	1-3/4	1-7/8
Distance floor to table, range, inches	25 to 36	25 to 36	25 to 36
Motor horsepower, minimum	1/2	1/2	1

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TABLE I - Size and capacities. cont.

Characteristics	Size 18	Size 20	Size 24
Capacity (in mild steel), inches	3/4	1	1-1/4
Centerline of spindle to column face, inches	9	10	12
Swing, inches, Style A	18	20	24
Overhang, inches, Style B through F	9	10	12
Sliding head adjustment, inches	5	5	6
Spindle traverse, inches	5	5	5-3/4
Spindle nose to table maximum inches	25	25	25
Spindle speeds, number	Infinite	Infinite	Infinite
Spindle speed range, revolution per minute	400 to 3000	200 to 2000	150 to 1500
Spindle diameter, (at drive end), inches	7/8	1	1-1/8
Quill diameter, inches	1-7/8	2	2-3/8
Distance floor to table, range, inches	25 to 36	25 to 36	25 to 36
Motor horsepower, minimum	2	3	5

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TABLE II. Alignment and accuracy tolerance.

Characteristics	Maximum allowable deviation
Working surface of worktable, flat	.0016"/12"
Column guideways square with working surface of the worktable, in longitudinal and traverse planes	.003"/12"
Spindle taper, runout 1 inch from nose	.001" TIR
Spindle taper, runout 6 inches from nose	.002" TIR
Spindle square with working surface of the worktable, turn around method with 6-inch radius, in longitudinal and transverse planes	.003"/12"

Custodians:
Army - AL
Navy - SH
Air Force - 99

Review activities:
Navy - SH
Air Force - 84

User activities:
Army - SM
Navy - YD, MC

Preparing activity:
DLA - IP

Project (3413-0148)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-D-80076D	2. DOCUMENT DATE (YYMMDD) 901001

3. DOCUMENT TITLE: Drilling Machine, Upright, Box Column, Floor Type Plain Table, Hand Feed

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
Defense Industrial Plant Equipment Center ATTN: DIPEC-SSM	(1) Commercial 901-775-6257	(2) AUTOVON 683-6257
c. ADDRESS (Include Zip Code) 2163 Airways Blvd. Memphis, TN 38114-5051	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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